

Quantum Mpemba Effect in Disordered Lattices

P. E. Chuklanov¹, M. A. Less¹, P. V. Zakharenko², D. E. Zaitsev¹,
A. P. Alodjants^{2*}

¹Moscow Institute of Physics and Technology, Dolgoprudny, Russia.

²ITMO University, Saint Petersburg, Russia.

*Corresponding author(s). E-mail(s): Alexander.ap@list.ru;

Abstract

We investigate the Quantum Mpemba effect (QME) in which a quantum state initially farther from equilibrium relaxes faster than one starting closer. We consider single-photon dynamics in networks of coupled optical waveguides to examine QME. The system is modeled as a quantum walk on a graph subject to local dephasing within the Lindblad master equation framework. By spectrally decomposing the Liouvillian superoperator for regular square lattices, we identify initial states with selective mode content that exhibit pronounced anomalous relaxation. We then introduce a Watts–Strogatz-type edge-rewiring procedure and show that topological disorder localizes the slowest relaxation mode while higher-order modes remain delocalized. This localization asymmetry gives rise to a natural separation of network nodes into “bright” and “dark” regions with respect to the fundamental mode, enabling a purely geometric construction of Mpemba state pairs. Based on this finding, we propose a deterministic heuristic algorithm that generates initial states with a pronounced Mpemba effect for selected disordered lattice topologies and validate it against Monte Carlo sampling.

Keywords: Quantum Mpemba effect, Disordered lattices, Open quantum systems, Quantum walks

1 Introduction

The Mpemba effect is a counterintuitive relaxation phenomenon in which a system initially farther from equilibrium reaches its stationary state faster than one that starts closer. It was first systematically described by Mpemba and Osborne in 1969 for hot water freezing faster than cold [1]. Subsequent studies revealed that this effect

is not limited to thermal systems but can be understood more generally through the spectral structure of relaxation dynamics — a framework established for Markovian systems by Lu and Raz [2] and sharpened by the strong-Mpemba classification of Klich *et al.* [3]. Within this framework, anomalously fast relaxation occurs when the initial state has a weak overlap with the slowest decaying eigenmode of the dynamical generator; the relaxation is then governed primarily by faster modes, and the system approaches equilibrium more quickly. This mechanism has been confirmed experimentally in classical colloidal systems, where accelerated cooling was observed under controlled conditions [4].

In quantum physics, the Mpemba effect acquires a richer structure (see [5] for a recent review). Transplanting the classical spectral mechanism [2, 3] to the quantum domain, Carollo, Lasanta, and Lesanovsky [6] showed that in Markovian open quantum systems governed by a Lindblad master equation, an appropriately engineered unitary rotation of the initial state can render it orthogonal to the slowest Liouvillian eigenmode, thereby *exponentially* accelerating the approach to stationarity. This strong form of the QME relies on the spectral decomposition of the Liouvillian superoperator: if the overlap coefficient c_1 with the first non-trivial right eigenvector vanishes, the effective relaxation gap is set by the second eigenvalue and convergence becomes exponentially faster. In [7], the authors subsequently demonstrated that global spin rotations can serve as a practical, albeit approximate, implementation of this idea in dissipative spin lattices, providing a physically realizable protocol for accelerated relaxation.

The QME has since been predicted and observed across a wide range of physical platforms. In isolated many-body quantum systems, an analogous effect manifests through anomalous symmetry restoration: a tilted ferromagnet with *greater* initial symmetry breaking can restore its symmetry *faster* than one closer to the symmetric state, as probed via entanglement asymmetry [8]. The experimental observation of this phenomenon was reported by Joshi *et al.* [9] using a trapped-ion quantum simulator, where the effect was monitored through randomized measurements and classical shadows. In the domain of open quantum systems, the strong QME—characterized by an exponentially accelerated approach to the steady state—was experimentally demonstrated in [10] on a single trapped-ion qubit. In that work, the strong Mpemba condition was shown to coincide with a Liouvillian exceptional point, revealing a deep connection between anomalous relaxation and non-Hermitian physics. The thermodynamic content of the effect has likewise been analyzed in mesoscopic transport settings [11] and through the lens of work and entropy production [12]. Photonic lattices provide a particularly natural and experimentally accessible arena for studying the QME. In particular, Longhi [13] demonstrated that the Mpemba effect arises in the diffusive dynamics of light in finite-sized photonic lattices under dephasing: certain highly localized initial intensity distributions reach the uniform (equilibrium) distribution faster than initially delocalized ones. Extending these results to the genuinely quantum-optical regime, Longhi [14] predicted a *bosonic* Mpemba effect in leaky optical resonators and waveguides by exploiting non-classical states of light—Fock states, coherent states, and squeezed states—whose distinct photon statistics lead to qualitatively different relaxation pathways. Together, these photonic studies establish coupled

waveguide networks as a versatile and tunable platform for engineering anomalous relaxation phenomena, bridging the gap between abstract spectral theory and concrete quantum-optical implementations.

In this work, we investigate the QME on graphs that model networks of coupled optical waveguides. We consider the quantum walk of a single photon with local dephasing described by a Lindblad master equation. The central idea is to exploit the spatial structure of the Liouvillian eigenmodes—specifically, the distinction between “bright” modes (efficiently excited by local perturbations) and “dark” modes (spatially decoupled from most of the network)—to engineer initial states with controlled overlap with the slowest relaxation channels. The role of network topology, and in particular the transition from regular to disordered (small-world) lattices via an edge-rewiring procedure, is a key element of our analysis.

The paper is organized as follows. In Sec. 2, we introduce the model: the tight-binding Hamiltonian for single-photon walks on graphs, the Lindblad master equation with local dephasing dissipators, and the distance metric based on quantum relative entropy used to identify the Mpemba effect. In Sec. 3, we present the Liouvillian vectorization and spectral decomposition formalism. In Sec. 4, we analyze the mode structure of regular square lattices, constructing spatial population maps and identifying geometric regularities. In Sec. 5, we use the spectral decomposition to design candidate initial states with selective mode content and verify the Mpemba effect through numerical simulation. In Sec. 6, we describe the Watts–Strogatz-type edge-rewiring procedure that interpolates between regular and random topologies, and characterize the associated small-world crossover. In Sec. 7, we perform a detailed spectral and eigenvector analysis of the disordered lattices, tracking the evolution of relaxation rates, the Inverse Participation Ratio, and the community overlap of the slowest modes as functions of the rewiring probability. We introduce the concept of “Topological Mpemba Effect” and formulate the hypothesis on the bright-to-dark transition of the slowest mode. In Sec. 8, we leverage the local excitability maps to propose a deterministic heuristic algorithm for constructing pairs of initial states that exhibit a pronounced Mpemba effect on selected network topologies, and validate it against Monte Carlo sampling. Finally, Sec. 9 summarizes the main results and discusses perspectives for future work.

2 Model

2.1 Hamiltonian of Single-Photon Walk on a Graph

We consider the quantum walk of a single photon in a network of coupled optical waveguides [15]. In such arrays the role of time is played by the propagation distance along the waveguides, so all temporal dynamics below translates directly into evolution along the sample [13, 15]. In the tight-binding approximation, the dynamics of the system are governed by the Hamiltonian:

$$\hat{H} = -\frac{J}{2} \sum_{i,j} \tau_{ij} (\hat{a}_i^\dagger \hat{a}_j + \hat{a}_j^\dagger \hat{a}_i), \quad (1)$$

where $\hat{a}_i^\dagger$ ($\hat{a}_i$) is the photon creation (annihilation) operator at site i , J is the connection constant (hopping rate), which is the same for all nodes, and τ is the symmetric adjacency matrix of the N -node graph.

To include environmental decoherence into our model, we employ the Lindblad master equation framework [16–18]. For a network of optical waveguides, the dominant source of decoherence is local dephasing, which is described by the local number operators acting as dissipators, $\hat{L}_k = \hat{a}_k^\dagger \hat{a}_k = \hat{n}_k$. Coherent hopping supplemented by local dephasing places the model in the class of quantum stochastic walks, which interpolate between coherent quantum and classical diffusive transport [19]. Physically, pure dephasing captures the random propagation-constant fluctuations induced by fabrication disorder and environmental noise, which destroy inter-site phase coherence while conserving the photon number; this is the dominant decoherence channel considered in photonic realizations of anomalous relaxation [13]. Uniform propagation losses — the other ubiquitous imperfection — factor out of the single-photon sector as a global attenuation and leave the *normalized* relaxation dynamics, and hence all trajectory crossings studied below, unchanged; site-resolved loss engineering is beyond the scope of this work. The evolution of the system’s density matrix $\hat{\rho}$ is given by:

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}, \hat{\rho}] + \gamma \sum_{k=1}^N \left(\hat{L}_k \hat{\rho} \hat{L}_k^\dagger - \frac{1}{2} \{ \hat{L}_k^\dagger \hat{L}_k, \hat{\rho} \} \right), \quad (2)$$

where $\gamma > 0$ is the dephasing rate.

We restrict our analysis strictly to the single-photon subspace, spanned by the localized basis states $|k\rangle$. In this basis, the number operator acts as a projector, $\hat{n}_k|m\rangle = \delta_{k,m}|m\rangle$. Evaluating the matrix elements $\rho_{k,l} = \langle k|\hat{\rho}|l\rangle$, the coherent part naturally yields $H_{k,m} = -J\tau_{k,m}$ (considering that the adjacency matrix is symmetric due to the non-directionality of the graph). The dissipative part simplifies significantly since $\langle k|\hat{n}_m \hat{\rho} \hat{n}_m|l\rangle = \delta_{k,m} \delta_{l,m} \rho_{k,l}$. Summing over all sites m , the full evolution equation reduces to the explicit element-wise form [13]:

$$\frac{d\rho_{k,l}}{dt} = iJ \sum_{m=1}^N (\tau_{k,m} \rho_{m,l} - \rho_{k,m} \tau_{m,l}) - \gamma(1 - \delta_{k,l})\rho_{k,l}. \quad (3)$$

As Eq. (3) demonstrates, the dissipative term causes the off-diagonal coherences ($\rho_{k,l}$ for $k \neq l$) to decay at rate γ , while the diagonal populations ($\rho_{k,k}$) remain unchanged in the absence of hopping.

2.2 Distance Metric and the Mpemba Effect

The essence of the Quantum Mpemba effect lies in anomalous relaxation dynamics: a state initially further from equilibrium reaches it faster than a state initially closer. To quantify this phenomenon, we require a rigorous geometric distance metric between the state $\hat{\rho}(t)$ and the stationary state $\hat{\rho}^{(S)}$.

Following standard information-theoretic approaches to open quantum systems, we measure the distance to equilibrium using the quantum relative entropy:

$$D(\hat{\rho}(t)||\hat{\rho}^{(S)}) = \text{Tr}\{\hat{\rho}(t) \log \hat{\rho}(t)\} - \text{Tr}\{\hat{\rho}(t) \log \hat{\rho}^{(S)}\}. \quad (4)$$

We select the quantum relative entropy because it is a fundamental measure of distinguishability between quantum states and is monotonically non-increasing under completely positive trace-preserving (CPTP) maps [20]. It represents the quantum generalization of the classical Kullback-Leibler divergence. While the choice of metric is not uniquely restricted, previous studies indicate that the Quantum Mpemba Effect is a robust dynamical feature qualitatively observable with alternative measures, such as the trace distance or the Hilbert-Schmidt distance [6].

For our model, the stationary state is the maximally mixed state within the single-photon subspace, corresponding to a uniform probability distribution across all N graph nodes: $\hat{\rho}^{(S)} = \frac{1}{N}\hat{I}$. Given that $\text{Tr}(\hat{\rho}) = 1$, substituting this into the relative entropy definition conveniently simplifies the expression to:

$$D(\rho(t)) = \log(N) + \text{Tr}\{\hat{\rho}(t) \log \hat{\rho}(t)\}. \quad (5)$$

Here, the second term is the negative von Neumann entropy of the state [21]. Thus, for the dynamics considered here, the relative entropy and the von Neumann entropy-based distance are mathematically equivalent up to a constant shift. An intersection of the $D(\rho(t))$ trajectories for two distinct initial states A and B —that is, $D_A(0) > D_B(0)$ but $D_A(t) < D_B(t)$ at some later time $t > 0$ —serves as the unambiguous hallmark of the Mpemba effect.

2.3 Liouvillian Vectorization

To solve Eq. (3) and analyze the system's spectrum, we transition to the Liouville superoperator formalism [22]. The density matrix is mapped to a column vector via the vectorization operation $\text{vec}(\cdot)$, denoted in double-ket notation as $|\rho\rangle\rangle \equiv \text{vec}(\rho) = [\rho_{1,1}, \rho_{2,1}, \dots, \rho_{N,N}]^\top$. We further define the inverse operator $\text{Mat} \equiv \text{vec}^{-1}$, which restores an $N \times N$ matrix from its vectorized form. The Liouville space is equipped with the Hilbert-Schmidt inner product $\langle\langle A|B\rangle\rangle = \text{Tr}(A^\dagger B)$.

The master equation transforms into a matrix differential equation $\frac{d}{dt} |\rho\rangle\rangle = L |\rho\rangle\rangle$, where the matrix representation L of the Liouvillian superoperator $\mathcal{L}$ has dimensions $N^2 \times N^2$ and can be explicitly constructed by using the Kronecker product ($\otimes$). Separating the coherent and dissipative parts, we obtain the target matrix:

$$L = iJ(I \otimes \tau - \tau \otimes I) + L_D, \quad (6)$$

where I is the $N \times N$ identity matrix, and L_D is a diagonal matrix with elements equal to 0 for components corresponding to populations ($\rho_{k,k}$) and $-\gamma$ for coherences ($\rho_{k,l}$).

Throughout the subsequent numerical analysis, we define the time scale in units of the inverse hopping rate, effectively setting $J = \frac{1}{2}$ (all times below are quoted in these dimensionless units; in a waveguide implementation they correspond to propagation

distance measured in units of the inverse coupling constant). The dephasing rate is fixed at $\gamma = 0.1$. This specific ratio ($\gamma/J = 0.2$) places the system in a weakly dissipative regime where coherent dynamics remains appreciable on the relaxation time scale.

3 Eigenstate Decomposition

As established, the dynamics equation is written as $\frac{d}{dt} |\rho\rangle\rangle = L |\rho\rangle\rangle$. This implies that its solution can be represented as:

$$|\rho(t)\rangle\rangle = e^{Lt} \cdot |\rho(0)\rangle\rangle \quad (7)$$

We analyze the system in the non-Hermitian diagonal basis of the operator L .¹ This vectorization formalism allows for the spectral analysis of the Liouvillian superoperator.[23]

To spectrally decompose the dynamics, we introduce a biorthogonal basis consisting of the right eigenvectors v_k ($|\rho_k\rangle\rangle$) and the left eigenvectors $\langle\langle\sigma_k|$, which satisfy:

$$L \cdot v_k = \lambda_k \cdot v_k, \quad \langle\langle\sigma_j|\rho_k\rangle\rangle = \mathcal{N}_k \delta_{jk}, \quad (8)$$

where $\mathcal{N}_k$ is a complex normalization constant. While analytical treatments often set $\mathcal{N}_k = 1$, standard numerical eigensolvers normalize left and right eigenvectors independently. Therefore, we explicitly retain the factor $\mathcal{N}_k = \langle\langle\sigma_k|\rho_k\rangle\rangle$ in all subsequent projections.

$$|\rho(0)\rangle\rangle = \sum_k c_k v_k \quad (9)$$

Substituting this decomposition into the dynamics formula yields:

$$|\rho(t)\rangle\rangle = \sum_k c_k e^{\lambda_k t} v_k \quad (10)$$

We uniquely identify the stationary state of the system corresponding to $\lambda_0 = 0$. The associated right eigenvector is the stationary density matrix $v_0 = |\rho^{(S)}\rangle\rangle$, and the corresponding left eigenvector is the identity operator. Due to the trace-preserving nature of the Lindblad evolution, the coefficient for the stationary mode is strictly $c_0 = 1$, since $\langle\langle\sigma_0|\rho\rangle\rangle = \text{Tr} \rho = 1$ both for the initial and for the stationary state.

Furthermore, $e^{\text{Re} \lambda_k t}$ represents the dissipative exponent in the decomposition; thus, $\text{Re} \lambda_k < 0$ for non-trivial modes, corresponding to the decay rate. The term $e^{i \text{Im} \lambda_k t}$ corresponds to the oscillation frequency $\omega_k = \text{Im} \lambda_k$. [23, 24]

¹We assume that the Liouvillian superoperator L is diagonalizable, which guarantees the existence of a complete biorthogonal basis of eigenmodes. While the non-Hermitian nature of L allows for the existence of non-diagonalizable defective degeneracies (Exceptional Points), such points typically form a manifold of measure zero in the parameter space and are avoided in our generic parameter regime. Furthermore, the geometric symmetries of the square lattice induce standard (semi-simple) macroscopic degeneracies in the spectrum—such as the observed quadruplets of modes with identical complex decay rates—which our numerical solver naturally resolves by identifying a linearly independent basis spanning the degenerate subspaces.

4 Lattice Investigation

We proceed with the investigation of the system using a small **square** lattice.

To this end, we construct mode population maps. The mode k , corresponding to the eigenvector v_k , is reshaped into matrix form via the Mat operator defined in Sec. 2.3. It is crucial to emphasize that $\text{Mat}(v_k)$ does not represent a physical state, as it is not positive semi-definite and its trace is generally non-unity (or zero for decaying modes). Then, the populations (diagonal components) are extracted from the resulting matrix; each such component corresponds to a single lattice site:

$$\vec{p}_k = \text{Re}(\text{diag}(\text{Mat}(v_k))). \quad (11)$$

The components of $\vec{p}_k$ are then mapped onto the **nodes of the original lattice**. In the diagram, color and brightness are normalized across the entire plot and illustrate the sign and magnitude, respectively.

Because the ideal square lattice possesses fundamental geometric symmetries (the D_4 point group), the Liouvillian superoperator commutes with the corresponding spatial parity operators. Consequently, the Liouvillian spectrum exhibits macroscopic degeneracies, with modes frequently emerging in multiplets (e.g., quadruplets) sharing identical decay rates $\text{Re}(\lambda)$. Crucially, rather than returning arbitrary linear combinations within these degenerate subspaces, the numerical eigensolver resolves the simultaneous eigenstates of the Liouvillian and the spatial parity operators. This yields a deterministic, orthogonal basis composed of distinct macroscopic patterns with well-defined spatial parities (e.g., distinctly horizontal or vertical boundary modes).

To systematically present these results without redundancy, the modes are organized into macro-groups k' sharing identical relaxation rates. Fig. 1 illustrates this symmetry-resolved basis for a 10×10 lattice.

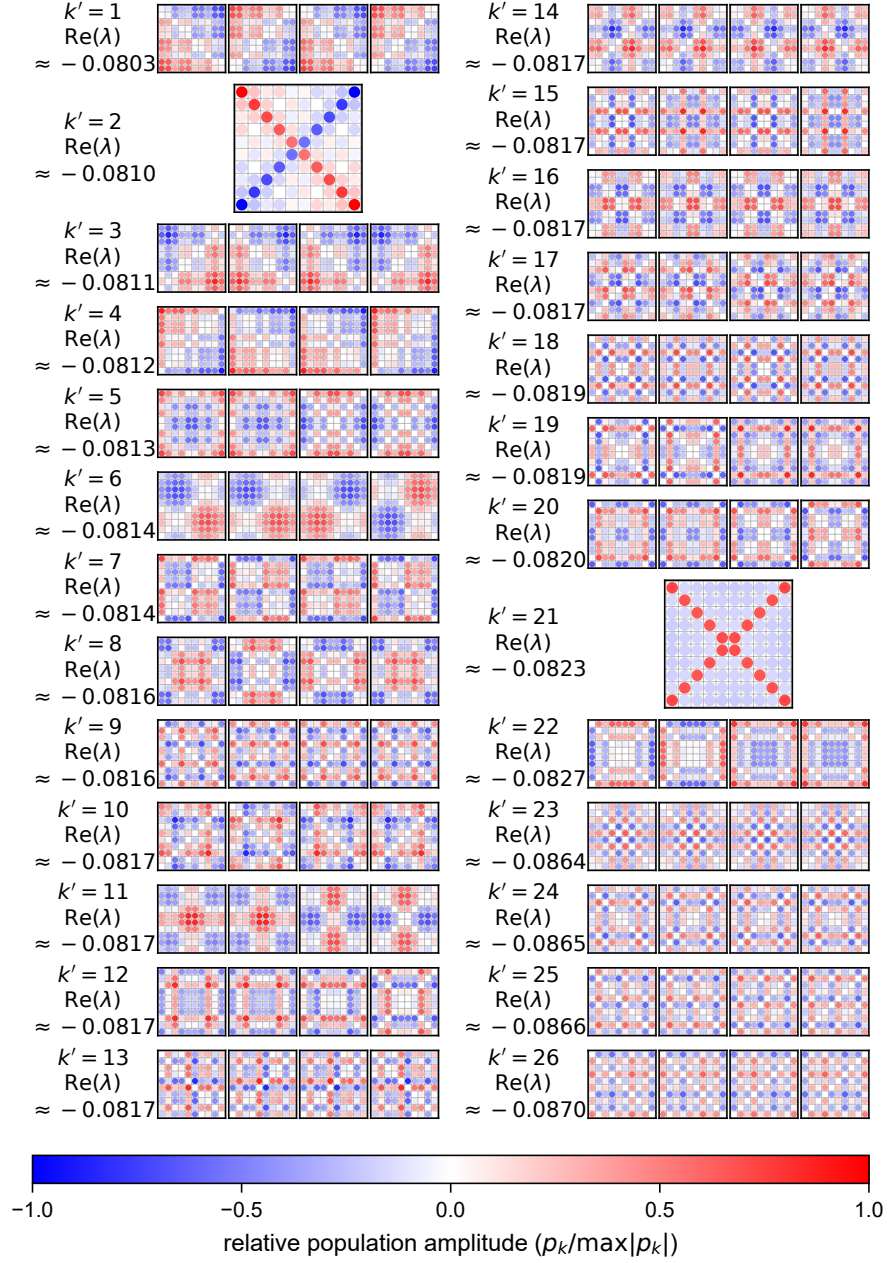


Fig. 1: Node-wise distribution of Liouvillian eigenmode populations for a 10×10 regular lattice. The maps represent the real part of the diagonal components of the reshaped eigenvectors $\vec{p}_k$ (eq. 11). Modes are organized into groups k' sharing identical decay rates ($\text{Re}(\lambda)$); for example, the group $k' = 1$ corresponds to the degenerate subspace containing modes $k = 1, \dots, 4$ due to lattice symmetry. Color intensity of each node is normalized globally by the maximum absolute population value, $\max |p_k|$, found among all non-trivial modes. Red denotes positive values, blue indicates negative values, with opaque colors corresponding to a relative amplitude of ± 1 (i.e., $\pm 100\%$ of $\max |p_k|$).

5 Analysis of Mode Structure

An extensive analysis of the spatial mode distributions reveals distinct geometric regularities, as exemplified by the 10×10 lattice in Fig. 1. Rather than resembling random noise, the slow modes exhibit clear, macroscopic patterns governed by lattice symmetries. Examining the figure, we immediately observe that the modes in the slowest degenerate subspace ($k' = 1$) are strongly localized at the lattice boundaries and corners. As the decay rate increases, these macroscopic features shift, giving rise to pronounced diagonal and anti-diagonal structural motifs (e.g., $k' = 2$ and $k' = 21$). Conversely, as the mode index increases further, the spatial distributions become less globally expressive. The higher-order modes lose their macroscopic structure, breaking down into fine-grained, small-scale spatial fluctuations (high spatial frequencies) that eventually resemble unstructured noise.

This clear structural hierarchy—from macroscopic, boundary-localized patterns to fine-scale local fluctuations—provides a powerful heuristic principle for investigating the Quantum Mpemba effect.² Initial states engineered to geometrically match the macroscopic "footprint" of specific slow modes are highly likely to excite those modes with extreme selectivity. Based on this physical insight, we empirically formulated a set of eight distinct initial states. These states serve as our "candidates" for demonstrating anomalous relaxation, as they are explicitly designed to couple predominantly into only a handful of targeted relaxation channels.

The empirical candidate states for a square lattice are defined as follows:

1. **Pure (Center):** A photon localized at a single central site (for even N , any of the four central sites), probing modes with central density.
2. **Mixed (Two Corners):** An equal mixture of states at two opposite corners, *targeting the symmetries seen in $k' = 1, 4, 5, 7, 22$ in Fig. 1.*
3. **Mixed (Four Corners):** A statistical mixture of all four corner sites.
4. **Mixed (Diagonal):** An incoherent mixture of sites along the main diagonal, *subspaces like $k' = 2, 21$ in Fig. 1.*
5. **Pure (Coherent Diagonal):** A coherent superposition supported entirely on the geometric main diagonal: $|\psi\rangle = \frac{1}{\sqrt{|S|}} \sum_{j \in S} |j\rangle$ with $S = \{(i, j) : i = j\}$.
6. **Mixed (Inner Corners):** A mixture of the four inner corners $\{(2, 2), (n - 1, 2), (2, n - 1), (n - 1, n - 1)\}$.
7. **Mixed (Top/Bottom Edges):** A state distributed evenly along the top and bottom boundary rows. *E.g. $k' = 2, 3, 4, 7$ in Fig. 1*
8. **Mixed (Checkerboard):** A macroscopic checkerboard pattern (e.g., only "even" nodes occupied), *specifically designed to interact with high-frequency modes similar to $k' = 26$ in Fig. 1.*

To verify our heuristic selection, we quantitatively evaluate the magnitude of the mode contribution to each state by computing the projection onto the corresponding

²As defined in Sec. 2.2, the Quantum Mpemba effect occurs when a state with a larger initial distance $D(\rho(0))$ relaxes to the stationary state faster than one with a smaller initial distance, leading to a crossing in their distance trajectories.

Liouvillian eigenbasis:

$$c_k = \frac{\langle\langle\sigma_k|\rho(0) - \rho^{(S)}\rangle\rangle}{\langle\langle\sigma_k|\rho_k\rangle\rangle} \quad (12)$$

Where: $|\rho(0)\rangle\rangle$ — vectorized initial density matrix. $|\rho^{(S)}\rangle\rangle$ — vectorized stationary state density matrix. $|\rho_k\rangle\rangle$ — k -th right eigenvector of the Liouvillian (v_k). $\langle\langle\sigma_k|$ — k -th left eigenvector of the Liouvillian ($w_k^\dagger$). The pairwise normalization by $\langle\langle\sigma_k|\rho_k\rangle\rangle$ is well defined only for non-degenerate eigenvalues. Within the degenerate multiplets of the regular lattice the numerically returned left and right eigenvectors are not biorthogonally paired, so in practice the coefficients are obtained as the exact solution of the linear system $\sum_k c_k |\rho_k\rangle\rangle = |\rho(0) - \rho^{(S)}\rangle\rangle$; the accuracy of every decomposition is monitored through its reconstruction residual ($\leq 10^{-10}$ for all eight states).

In the right diagram (Fig. 2), we plot the percentage contribution of each degenerate multiplet k' (the macro-groups introduced for Fig. 1), $W_{k'} = \|\sum_{k \in k'} c_k |\rho_k\rangle\rangle\|^2$, normalized over the non-stationary multiplets. Individual coefficients c_k within a degenerate multiplet depend on the arbitrary basis choice made by the numerical eigensolver, whereas the norm of the projected component is basis-invariant. The weights $W_{k'}$ do not possess a direct probabilistic interpretation. Instead, they serve as a relative geometric measure of how strongly a specific transient channel is excited by the initial configuration.

We decompose each of the 8 candidate initial states into the lattice eigenbasis to quantify their mode content. The projection coefficients W_k indicate how strongly a specific mode participates in the relaxation process of a given state. The results, summarized in Fig. 2, confirm that our empirically chosen states are indeed highly selective. The bar charts reveal a striking sparsity: the incoherent mixtures couple to a small number of multiplets rather than smearing across the spectrum. This is a direct consequence of **symmetry selection rules**: because these states exhibit distinct spatial symmetries (e.g., the fully symmetric "Four Corners"), their overlaps vanish for eigenmodes of mismatched spatial parity. The coherent superposition (state 5), which also populates the off-diagonal coherence sectors, couples to a broader yet still structured set of multiplets. This symmetry-driven selectivity is the fundamental prerequisite for observing the trajectory crossings characteristic of the Quantum Mpemba effect.

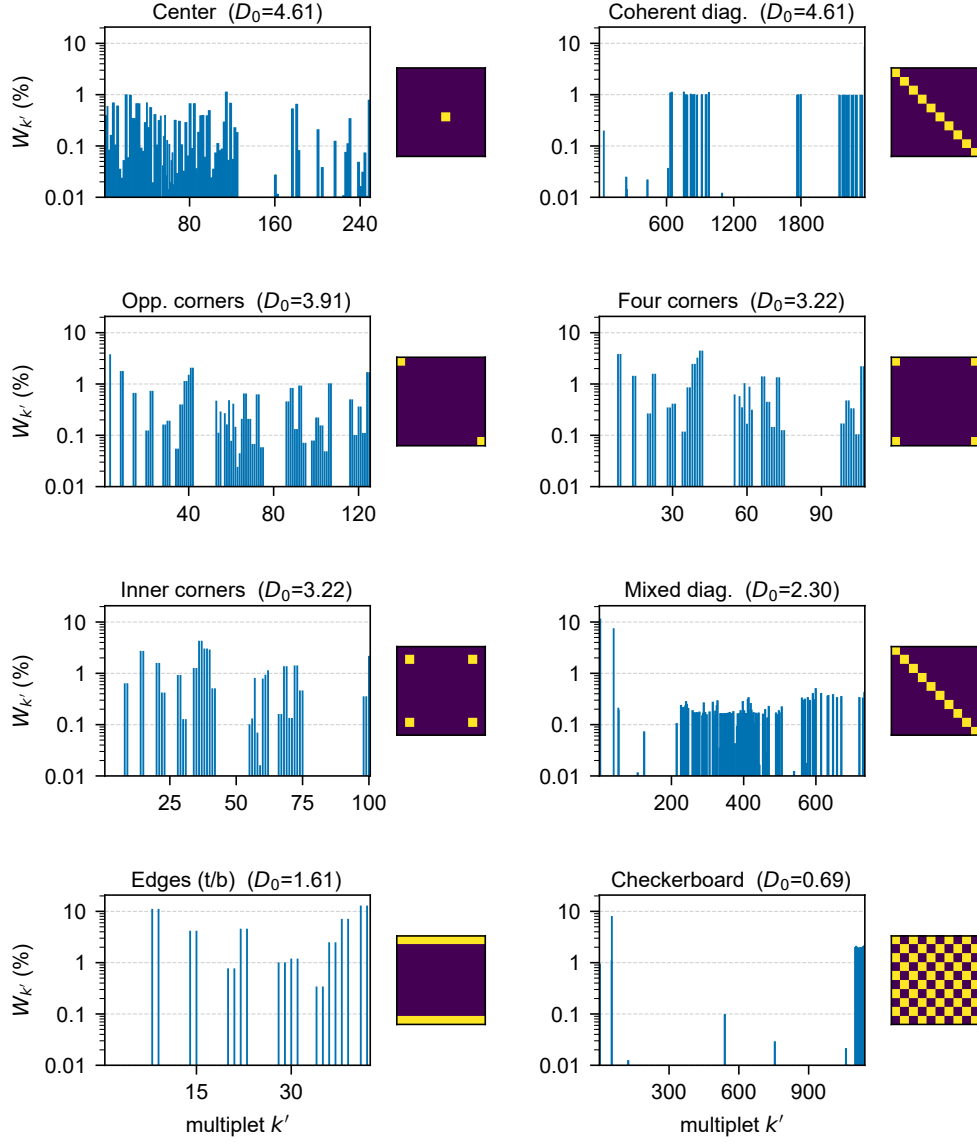


Fig. 2: Spectral decomposition of eight candidate initial states in a 10×10 lattice. Each **panel** analyzes a specific initial condition and consists of two elements: (1) a schematic illustration of the initial photon distribution (left), where yellow cells indicate nodes with a non-zero initial photon probability, serving as a binary visual guide independent of exact statistical weights or quantum phases; and (2) a bar chart of the multiplet contribution weights $W_{k'}$ — the squared norm of the state’s component within each degenerate multiplet k' (cf. eq. 12) — plotted on a **logarithmic scale** (right). Panels are ordered by decreasing initial distance to equilibrium $D_0 = D(\rho(0))$.

To avoid a misleadingly sparse appearance in linear scale (where the dominant bar masks the remaining weights), we present only the logarithmic visualization. The log scale highlights the intended point: the selected initial states have a strongly *mode-selective* decomposition while still exhibiting nonzero projections onto multiple faster modes.

5.1 Simulation

Subsequently, we performed a time-evolution simulation for these states to verify the existence of the predicted effect. As shown in Fig. 3, the trajectories for specific pairs cross, confirming the phenomenon. This behavior is robust and has been observed across a range of square lattice sizes ($n \in \{5, \dots, 11\}$).

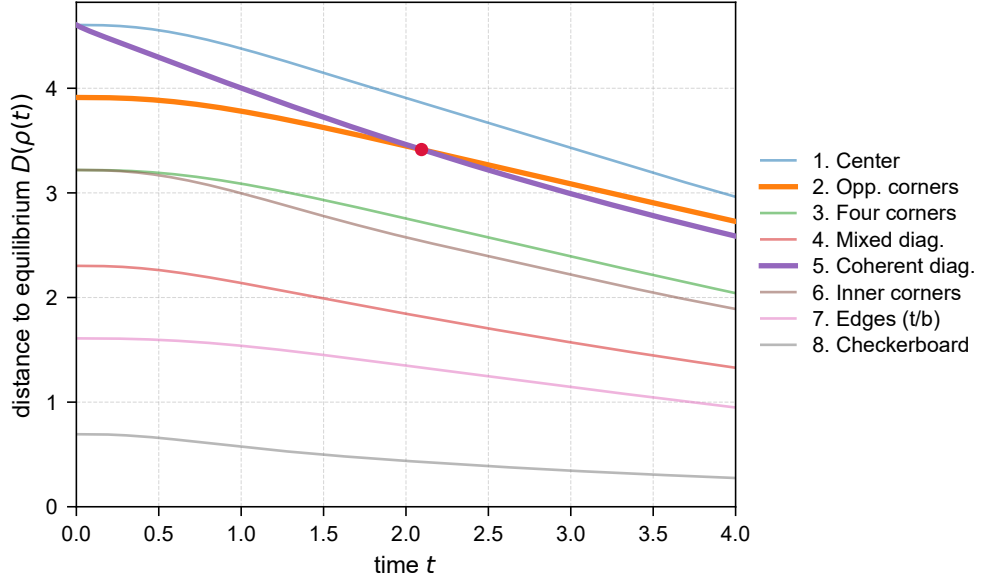


Fig. 3: Relaxation dynamics of the 10×10 lattice system shown as the time evolution of the distance to equilibrium $D(\rho(t))$ for eight distinct initial states (see Fig. 2). The crossing of trajectories (bold) represents the Mpemba effect, which is observed for the state pair (2 and 5).

6 Network Rewiring

We now modify the lattice structure by randomly rewiring a fraction of the edges. This process involves switching edge endpoints, thereby conserving the total number of edges in the system. The topology is governed by the parameter $p \in [0, 1]$, representing the rewiring probability for each initial edge. For an open $n \times n$ square lattice, the exact number of undirected edges is $E = 2n(n - 1)$. Consequently, the expected number of rewired edges is precisely $\langle E_{\text{rew}} \rangle = 2n(n - 1)p$.

In the context of open quantum systems, network fragmentation produces multiple degenerate stationary states—mirroring the classical correspondence between the multiplicity of the graph-Laplacian zero eigenvalue and the number of connected components [25]—which invalidates the spectral decomposition of the Liouvillian. Therefore, we must strictly enforce the global connectivity of the network. We achieve this through a Monte Carlo rejection sampling procedure, detailed in Algorithm 1.

Algorithm 1 Connectivity-Guaranteed Edge Rewiring

Require: Lattice dimension n , rewiring probability p , optional PRNG *seed*

Ensure: Connected adjacency matrix τ

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1: Initialize pseudo-random number generator with seed (if provided)
2: connected  $\leftarrow$  false
3: while not connected do
4:    $\tau \leftarrow \text{generate\_base\_grid}(n \times n)$ 
5:    $\mathcal{E} \leftarrow$  list of all edges in  $\tau$ ; Randomly shuffle  $\mathcal{E}$ 
6:   for each edge  $(u, v) \in \mathcal{E}$  do
7:     if  $\text{rand}(0, 1) < p$  then
8:       Select node  $X$  uniformly from  $\{u, v\}$ 
9:       Let  $Y$  be the other node:  $Y \leftarrow \{u, v\} \setminus \{X\}$ 
10:       $\mathcal{V}_{\text{valid}} \leftarrow$  all nodes  $\setminus (\{X, Y\} \cup \text{neighbors of } X \text{ in } \tau)$ 
11:      if  $\mathcal{V}_{\text{valid}} \neq \emptyset$  then
12:        Select target  $W$  uniformly from  $\mathcal{V}_{\text{valid}}$ 
13:        Remove edge  $(X, Y)$  and add edge  $(X, W)$  to  $\tau$ 
14:      end if
15:    end if
16:  end for
17:  if  $\text{check\_connectivity\_BFS}(\tau) == \text{true}$  then
18:    connected  $\leftarrow$  true
19:  end if
20: end while
21: return  $\tau$ 

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To prevent the creation of self-loops and multi-edges, target nodes are restricted to the set $\mathcal{V}_{\text{valid}}$. If no valid targets remain for a selected edge, its rewiring is simply bypassed (lines 11–14). Disconnected realizations are discarded via Breadth-First Search (BFS) validation (lines 17–19). This rejection step conditions the statistical ensemble on global connectivity: the acceptance fraction is essentially unity for $p \lesssim 0.05$, decreases to $\approx 86\%$ at $p = 0.3$, and reaches $\approx 20\%$ at $p = 1$, so ensemble averages at strong rewiring refer to the connected subensemble. Finally, to guarantee the exact reproducibility of the representative single-graph spatial maps presented in subsequent sections, the generator is initialized with a fixed seed. For statistical ensemble averaging, the seed varies to ensure independent sampling.

In the following sections, we focus on 10×10 lattices

6.1 Classical Network Analysis

A standard method for investigating phase transitions (or smoother crossovers) in networks involves analyzing the average shortest path length (averaged over all node pairs) as a function of the disorder parameter—in our case, the rewiring probability p [26]. For a more comprehensive analysis of such transitions and the “small-world” property, see [27, 28].

We plot this dependence for a 10×10 lattice across different scales. To faithfully reproduce the canonical Watts–Strogatz crossover [29] for this specific macroscopic metric, the strict BFS connectivity filter (Algorithm 1) was relaxed: graphs were rewired in a single pass without the rejection step, and the average shortest-path length of each realization was computed over its largest connected component (which coincides with the full graph whenever the realization is connected). The goal of this figure is to identify the crossover region in p that will later be compared with spectral changes:

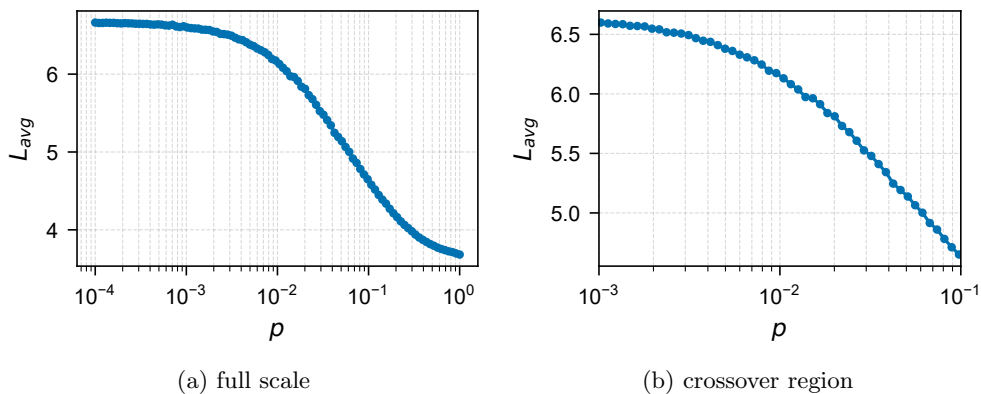


Fig. 4: Average shortest-path length versus rewiring probability p (in log. scale) for a 10×10 lattice, shown in two scalings. The smooth crossover indicates the onset of the small-world regime, i.e., the range of p where adding a small fraction of long-range links sharply reduces typical graph distances.

We observe a smooth crossover in the range $p \in [0.01, 0.1]$.

7 Mode Analysis for Disordered Lattices

We now turn to the analysis of disordered lattices obtained via the rewiring model described above, where the network topology is parameterized by the lattice size and the rewiring probability $p \in [0, 1]$. Our goal here is to understand how the spectral and spatial properties of the Liouvillian eigenmodes evolve as the network transitions from a regular lattice to a disordered graph.

7.1 Spectral Analysis

To establish a baseline for our spectral analysis, Fig. 5 presents representative Liouvillian eigenvalue spectra across several orders of magnitude of the rewiring probability p . This progression illustrates how increasing topological disorder breaks the geometric symmetries of the lattice and modifies the spectral structure near the origin, which governs the long-time relaxation dynamics of the system.

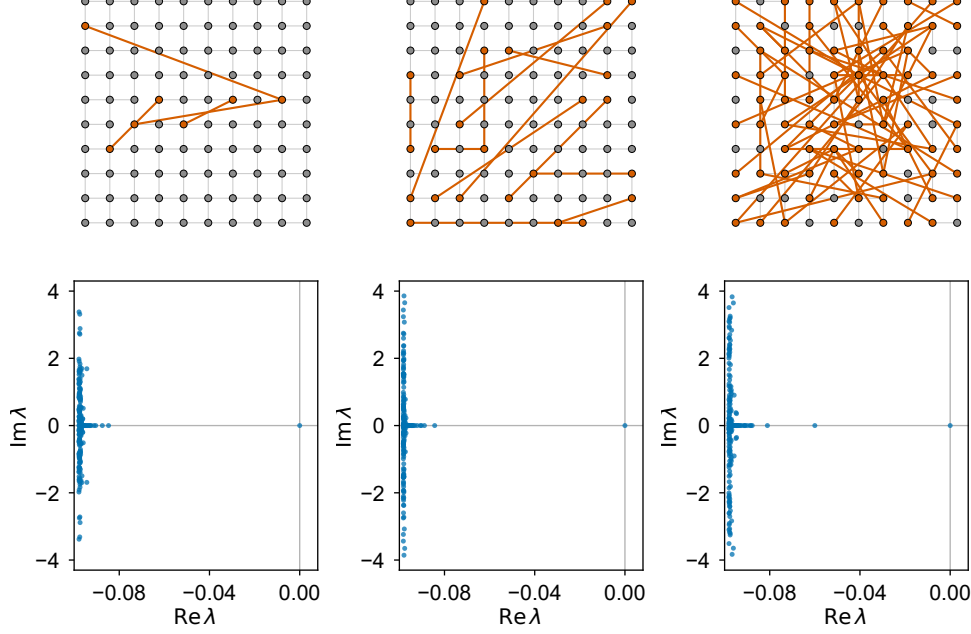


Fig. 5: Representative Liouvillian spectra for rewired 10×10 lattices at increasing disorder levels ($p = 0.01$, $p = 0.1$, $p = 0.3$). At weak disorder ($p = 0.01$) the spectrum retains the structured, near-degenerate branches inherited from the lattice symmetry; increasing the rewiring probability lifts these degeneracies and progressively smears the spectral branches. The eigenvalues closest to zero (rightmost points) correspond to the slowest relaxation modes.

7.2 Analysis of Slow Modes

We observe that the first few modes, located closest to zero on the real axis, are the most indicative of the system's long-term dynamics. Note that while these are conventionally the "slowest" decaying modes, they determine the convergence speed.

To investigate the system's response to disorder, for every disorder realization we identify the slowest non-trivial mode λ_1 as the eigenvalue with the largest real part. Depending on the realization, this mode is either purely real or belongs to an oscillating complex-conjugate family ($\text{Im } \lambda_1 \neq 0$) inherited from the degenerate multiplets of the

regular lattice. All statistics are collected over 30 independent graph realizations for each p ; the resulting relaxation rate is shown in Fig. 6(a).

For a detailed analysis across multiple orders of magnitude of p , Fig. 6 presents the mean relaxation rate of the slowest mode, the fraction of realizations in which this mode oscillates, the oscillation frequency of that subensemble, and the separation Δ between the two slowest *distinct* relaxation rates.

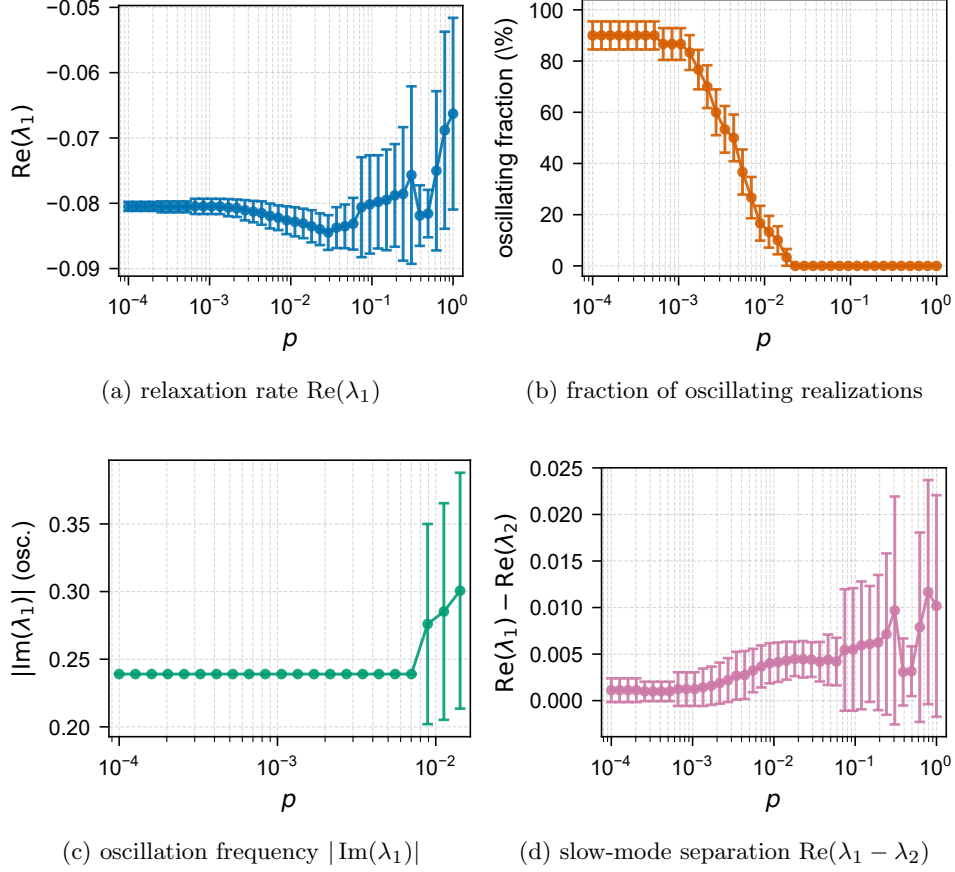


Fig. 6: Statistics of the slowest non-trivial Liouvillian mode versus the rewiring probability p (log scale) for a 10×10 lattice; for each realization the mode is identified by the largest $\text{Re } \lambda$. **(a)** Mean relaxation rate $\text{Re}(\lambda_1)$; the conventional Liouvillian relaxation gap is $-\text{Re}(\lambda_1)$. **(b)** Fraction of realizations whose slowest mode is oscillating ($\text{Im } \lambda_1 \neq 0$): the oscillating family inherited from the regular lattice loses its dominance with disorder and disappears for $p \gtrsim 2 \times 10^{-2}$. **(c)** Mean oscillation frequency $|\text{Im}(\lambda_1)|$ over the oscillating subensemble. **(d)** Separation between the two slowest *distinct* relaxation rates, $\Delta = \text{Re}(\lambda_1) - \text{Re}(\lambda_2)$; differences between individual modes are ill-defined inside degenerate multiplets. Error bars: ± 1 standard deviation over 30 realizations in (a), (c), (d); binomial standard error in (b).

We observe the following. First, for $p \lesssim 2 \times 10^{-2}$ the slowest mode of a given realization is predominantly a member of the oscillating family inherited from the regular lattice ($|\text{Im } \lambda_1| \approx 0.24$); disorder progressively suppresses this family, so the fraction of oscillating realizations drops below one half at $p \approx 4 \times 10^{-3}$ and vanishes for $p \gtrsim 2 \times 10^{-2}$ (panel b). Second, the mean relaxation rate is **non-monotonic**: $|\text{Re}(\lambda_1)|$ first grows with disorder — a mild ensemble-averaged *acceleration* of relaxation peaking at $p \approx 0.03$ — and then decreases toward the random-graph limit, corresponding to a **slowing down** of the slowest mode. The latter trend may appear **counter-intuitive**, since, on average, a small-world network relaxes to equilibrium faster; however, individual configurations hosting significantly slower localized modes emerge with increasing probability and dominate the ensemble average, which is accompanied by the sharp growth of the realization-to-realization scatter visible in panel (a). Rather than constituting the Mpemba effect itself—which strictly refers to the anomalous crossing of relaxation trajectories for specific initial states—this spectral phenomenon represents a **topology-induced slow-mode localization**. By structurally isolating the slowest relaxation channels, this topological transition geometrically assists in the preparation of initial states that bypass these slow channels, thereby facilitating what we term the **topologically assisted Quantum Mpemba effect**.

We confirmed the presence of a similar effect for different dimensions of the square lattice (not shown).

7.3 Eigenvector Analysis

We consider the Inverse Participation Ratio (IPR) parameter [30], which characterizes the degree of spatial localization of the eigenvectors. In the mode matrix $\text{Mat}(v_k)$, we retain only the diagonal components, as we are concerned solely with populations. Within (nearly) degenerate multiplets individual eigenvectors are defined only up to a basis rotation, so all localization measures below are evaluated for the distinct relaxation-rate groups k' introduced in Sec. 7.2: the diagonal weight of a group is computed from an orthonormal basis $\{q_k\}$ of the span of its eigenvectors, $d_i = \sum_k |(\text{Mat } q_k)_{ii}|^2$, a quantity invariant under intra-group basis rotations that reduces to $|(\text{Mat } v_k)_{ii}|^2$ for a non-degenerate mode. We then calculate:

$$IPR(k) = \frac{\sum_{i=1}^N |\rho_{kii}|^4}{(\sum_{i=1}^N |\rho_{kii}|^2)^2} \quad (13)$$

Here, the denominator serves as a normalization factor, ensuring that the IPR value ranges from $1/N$ (for a mode whose diagonal weight is uniformly distributed across all sites) to 1 (for a mode whose diagonal weight is concentrated at a single site). Consequently, a decrease in this parameter corresponds to a greater delocalization of the mode. To quantitatively trace the structural evolution of the states, we plot the average IPR for the three slowest distinct relaxation-rate groups as a function of the rewiring probability p , as shown in Fig. 7.

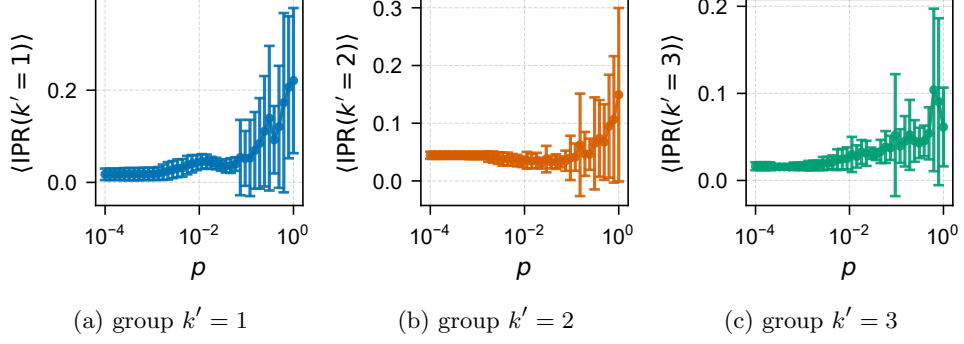


Fig. 7: Dependence of the IPR, defined in Eq. (13) with the invariant group weight d_i , on the rewiring probability p (log. scale) for the three slowest distinct relaxation-rate groups ($k' = 1, 2, 3$) in a 10×10 lattice. Error bars: ± 1 standard deviation over 30 network realizations. All three groups become more localized with increasing disorder; the leading group shows the strongest growth — from $\text{IPR} \approx 0.018$ (an extended state inherited from the lattice multiplets) to ≈ 0.22 at $p = 1$ — while the subordinate groups localize more weakly.

It is worth noting the distinct behavior of the leading group in Fig. 7: across the crossover region its IPR exceeds that of the subordinate groups by up to a factor of two, and the steep growth of its localization falls within the same small-world crossover region identified earlier.

7.4 Community Structure Analysis

We employ a more advanced method to analyze mode localization. The internal structure of each generated network is identified using the **Louvain method**, a popular and efficient algorithm [31]. It iteratively optimizes the modularity metric by grouping nodes to maximize the density of edges within communities and minimize edges between them. This results in a partition of all N nodes into M non-overlapping communities $C_1, C_2, \dots, C_M$.

For each Liouvillian eigenmode, we measure its concentration within the identified communities by introducing the **Overlap coefficient**. As established previously, the physical meaning of population distribution is carried by the diagonal of the reconstructed mode matrix $\text{Mat}(v_k)$. We define the “mass” of the mode at site i as $|\text{Mat}(v_k)_{ii}|^2$. The total diagonal “mass” of the mode, which serves as the normalization denominator in Eq. (15), is:

$$\text{TotalMass}(k) = \sum_{i=1}^N |\text{Mat}(v_k)_{ii}|^2 \quad (14)$$

The Overlap of mode k with community C_m is defined as the fraction of its total mass located within that community:

$$\text{Overlap}(k, C_m) = \frac{\sum_{i \in C_m} |\text{Mat}(v_k)_{ii}|^2}{\text{TotalMass}(k)} \in [0, 1] \quad (15)$$

- If $\max(\text{Overlap}(k)) \approx 1$, the mode is almost entirely contained within a single community. **It is structurally localized.**
- If $\max(\text{Overlap}(k))$ is small (e.g., close to $1/M$, where M is the number of communities), the mode is distributed across the entire network.

To observe the evolution of this structural localization, we calculate the maximum Overlap for the three slowest distinct relaxation-rate groups (using the invariant group weight d_i in place of $|\text{Mat}(v_k)_{ii}|^2$ for degenerate multiplets) across different rewiring probabilities p . The Louvain partitions are computed with a fixed random seed for reproducibility. The statistically averaged results are presented in Fig. 8.

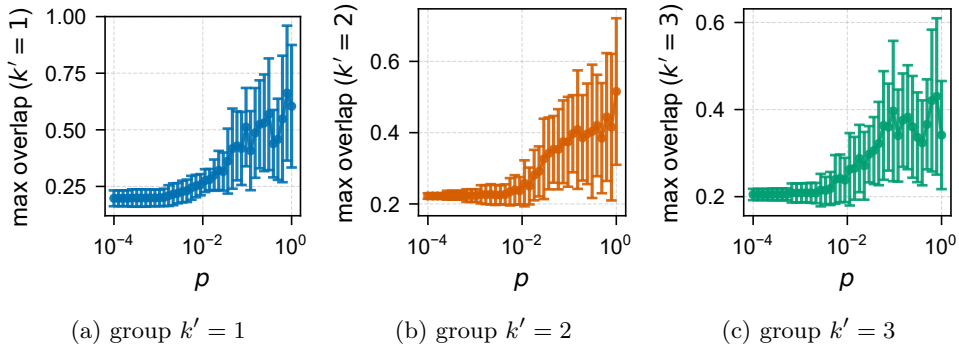


Fig. 8: Dependence of the maximum community Overlap coefficient, defined in Eq. (15), on the rewiring probability p for the three slowest distinct relaxation-rate groups ($k' = 1, 2, 3$) in a 10×10 lattice. Error bars: ± 1 standard deviation over 30 network realizations. The overlap of all three groups grows through the small-world crossover; the leading group ($k' = 1$) reaches the highest confinement (≈ 0.6), highlighting its structural localization within specific topological communities.

As depicted in Fig. 8, for the regular lattice structure ($p \rightarrow 0$), the slow-mode groups are distributed across several symmetry-related regions, and the community overlap is small (≈ 0.2). During the crossover, the network locally "breaks down" and connects to distant nodes while retaining local density inherited from the lattice, and the overlap of all three groups grows. For the subordinate groups ($k' = 2, 3$) the growth is moderate, and only the third group hints at a decline as $p \rightarrow 1$, within the realization scatter.

The leading group, however, stands out, indicating the re-emergence of the previously observed **topology-induced slow-mode localization**.

Its overlap grows monotonically and then reaches a plateau (≈ 0.6). Rather than being a strictly quantum anomaly, this behavior represents a fundamental feature of the Liouvillian dynamics in the present dephasing quantum-walk model, sharing deep structural similarities with the localization of slow relaxation modes (e.g., the Fiedler vector) in classical diffusion on heterogeneous networks. We observe that as p increases, despite the general delocalizing trend of the small-world crossover, **topological traps and distinct hubs emerge**; the modes localized around these structures "ascend" and become the slowest decay channels of the system.

The observed spatial localization of the slowest mode significantly affects the system's relaxation dynamics. According to the theory of the spectral Mpemba effect, a condition for accelerated relaxation is the ability to prepare an initial state that weakly excites (or "darkens") the slowest mode while strongly interacting with faster modes.

For delocalized modes (as in a regular lattice), such "darkening" requires complex interference tuning of amplitudes across the entire network. However, the localization of mode v_1 for $p > 0$ allows for the **geometric construction of excitation**:

if a mode is concentrated in a small region, the majority of network nodes become "dark" to it. To formalize this connection between topology and excitation selectivity, we must **shift from analyzing the internal mode structure to analyzing their response to local perturbations**.

7.5 Hypothesis on State Brightness

The brightness of a mode v_k is not determined by the mode alone; it is a **relational property** that characterizes the coupling of the mode to a specific class of initial states (or perturbations).

Since we work with the system's initial states, we select the class of fully **localized pure states** $\rho_i(0) = |i\rangle\langle i|$, where $i \in \{0, \dots, N-1\}$, as the fundamental basis of perturbations.

We then define **local excitability** $B(k, i)$ for a specific node i :

$$B(k, i) \equiv W_k(\text{for } |\rho_i(0)\rangle) = \left| \frac{\langle\langle \sigma_k | \rho_i(0) - \rho^{(S)} \rangle\rangle}{\langle\langle \sigma_k | \rho_k \rangle\rangle} \right|^2 \quad (16)$$

based on Eq. (12).

Since for $k > 0$ the left eigenvector $\langle\langle \sigma_k |$ is orthogonal to the right stationary state eigenvector $|\rho^{(S)}\rangle\rangle$ (i.e., $\langle\langle \sigma_k | \rho_0 \rangle\rangle = 0$), the formula for non-trivial modes simplifies to:

$$B(k, i) = \left| \frac{\langle\langle \sigma_k | \rho_i(0) \rangle\rangle}{\langle\langle \sigma_k | \rho_k \rangle\rangle} \right|^2, \quad \text{for } k > 0 \quad (17)$$

We define "dark" and "bright" modes as follows:

- A mode v_k is termed **globally bright** if its local excitability $B(k, i)$ is significant for the majority of sites i .

- A mode v_k is termed **geometrically dark** if its local excitability $B(k, i)$ is significant only for a small subset of sites and negligible for all others.

The previous results (increase in IPR and MaxOverlap for λ_1) allow us to formulate the following **hypothesis**:

- The slowest mode v_1 undergoes a transition from globally bright to geometrically dark for $p > 0$, as its spatial localization implies it can be efficiently excited only from a small region of the network.
- Faster modes (v_k for $k \geq 2$), remaining relatively delocalized, retain their globally bright character.

We verify this by constructing excitability maps for the modes of interest.

7.6 Visualization of Excitability Maps

As evidenced by the preceding analyses of the slow-mode separation (see Fig. 6), IPR (see Fig. 7), and community overlap (see Fig. 8), the system undergoes a pronounced structural crossover, with key metrics exhibiting extrema or changing their dependence trends within the $p \sim 0.1 - 0.2$ range. Consequently, we fix the rewiring probability at $p = 0.15$ for our subsequent spatial visualization, as it consistently captures this transition region where the topological localization of the slowest mode is most distinctly manifested.

To visualize the spatial structure, we construct a local excitability map by calculating the value $B(k, i)$ for each site i according to Eq. (16). Since the specific network topology is random for a fixed rewiring probability p , we generate a statistical ensemble of graph realizations. From this set, a “typical representative” is selected. As the selection criterion, we use the median average shortest path length L_{avg} of the ensemble. This specific metric is chosen not because of any *a priori* correlation with quantum relaxation dynamics, but because it is a fundamental macroscopic descriptor of networks. By targeting the L_{avg} , we ensure that the selected graph possesses a generic, statistically representative topology.

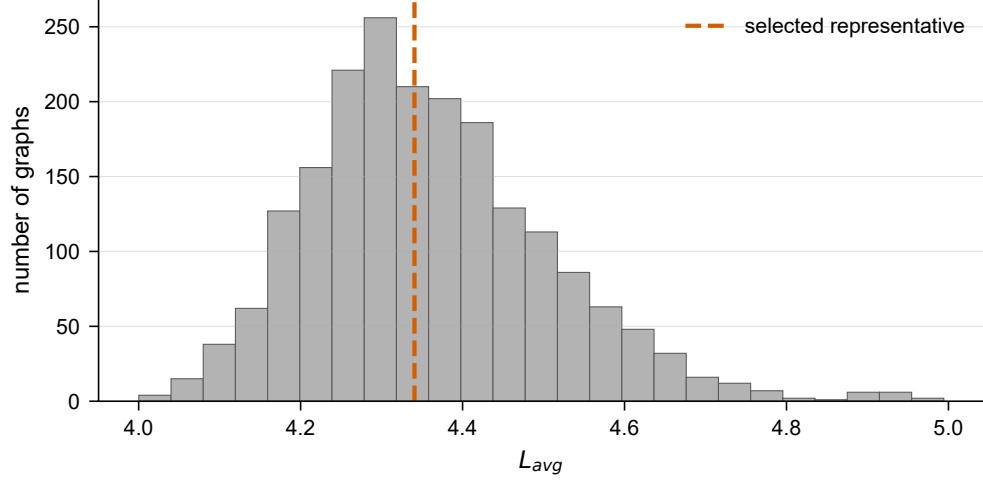


Fig. 9: Statistical distribution of the average shortest path length (L_{avg}) used to select a representative network topology at a fixed $p = 0.15$. The horizontal axis represents the topological metric L_{avg} , the vertical axis displays the frequency of occurrences across an ensemble of randomly generated graph realizations (ensemble size $N = 2000$). The red dashed vertical line marks the median of the distribution. Selecting the network realization closest to this median ensures that the subsequent spatial excitability analysis (Fig. 10) is performed on a typical architecture, thereby avoiding statistically anomalous configurations.

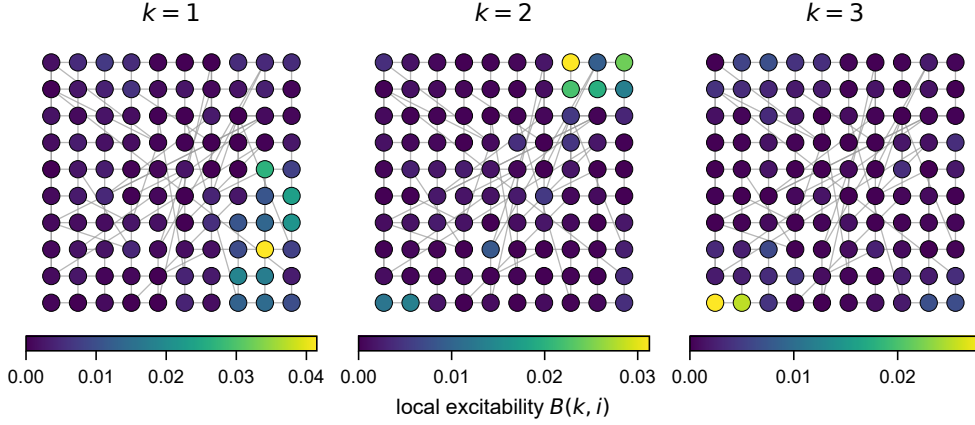


Fig. 10: Spatial maps of local excitability $B(k, i)$ (Eq. 17) for the first three non-trivial relaxation modes ($k = 1, 2, 3$) on a rewired 10×10 lattice ($p = 0.15$). Each lattice node represents a waveguide site. The color scale and marker size quantitatively encode the relative magnitude of the excitability: bright (yellow/red) nodes act as topological hubs that strongly couple to the respective mode upon local single-photon injection, whereas dark (purple/black) nodes represent topological voids that are effectively decoupled from the mode.

8 Heuristic Algorithm for Generating Initial States with a Pronounced Mpemba Effect

As established in the previous section, topological disorder can induce a pronounced spatial localization of the slowest relaxation mode ($k = 1$). This structural confinement physically implies that the fundamental mode becomes geometrically 'dark' to local excitations originating from the vast majority of the network nodes. Based on this observation, it is possible to bypass complex interference tuning and construct initial states with tailored relaxation rates using a purely geometric approach.

In this section, we propose a deterministic heuristic algorithm that leverages the local excitability map $B(1, i)$ to construct a pair of diagonal (incoherent) initial density matrices exhibiting a pronounced Quantum Mpemba Effect. We present this algorithm as a proof-of-concept demonstration. Rather than seeking the exact mathematical cancellation of the slowest mode projection ($c_1 \equiv 0$, which defines the strict 'Strong Mpemba' condition), our objective is to engineer a robust, physically observable and statistically exceptional effect. For illustrative purposes, all algorithmic steps and benchmarks are presented for a selected representative 6×6 network at a rewiring probability $p = 0.15$.

To examine the Mpemba effect, we refer to the state that is initially further from equilibrium as 'hot', since it relaxes more quickly, and to the state that is initially closer to equilibrium but relaxes more slowly as 'cold'.

8.1 Step 1: Spectral analysis

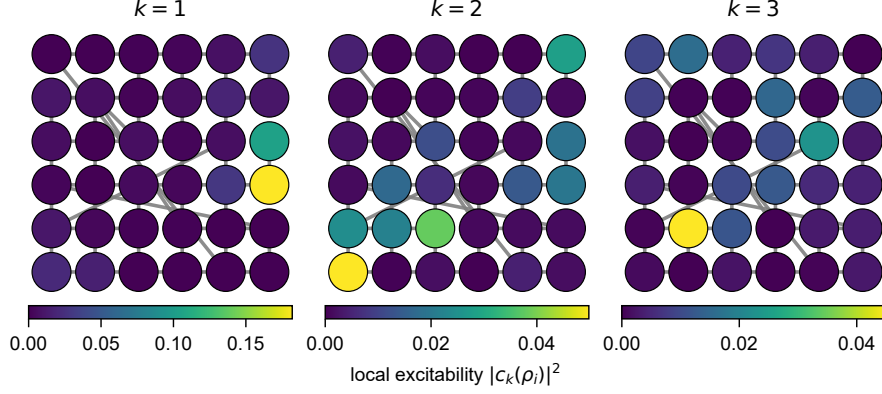


Fig. 11: Spatial maps of local excitability $B(k, i)$ for the first three non-trivial relaxation modes on the selected 6×6 test lattice ($p = 0.15$). The color mapping and node representation conventions strictly follow those detailed in Fig. 10. Note the pronounced spatial localization of the fundamental mode ($k = 1$), which provides the geometric basis for our state-engineering algorithm.

The foundation of our algorithm is the spatial inhomogeneity of the system's response to local injection. As derived in Eq. (16), the excitability metric $B(k, i)$ translates the abstract bi-orthogonal eigenvectors into a physical map, revealing how strongly a single photon injected at site i will excite the k -th relaxation channel.

Figure 11 illustrates these spatial excitability profiles. The maps reveal a critical structural feature: the fundamental mode ($k = 1$) exhibits extreme spatial localization, fragmenting the network into a few highly sensitive 'bright' hubs and a large 'dark' background of decoupled sites. Our state-engineering protocol exclusively relies on the excitability map of this slowest mode.

8.2 Step 2: Selection hot state (Dark node)

To construct the 'hot' state, we find the darkest node on the map (the node that the least excites the first, slowest mode) and inject a photon into this node. The density matrix of the 'hot' state is a matrix with a single unit on the diagonal and all other elements equal to zero.

8.3 Step 3: Selection cold state (Mix of bright nodes)

To construct the 'cold' state, we select an integer M ($2 \leq M \leq n^2/2$) and identify the M brightest nodes on the excitability map. The 'cold' state is then formed as an incoherent, equally weighted statistical mixture of these nodes: $\rho_{cold} = \frac{1}{M} \sum_{j \in \text{bright}} |j\rangle\langle j|$.

To determine the optimal number of bright nodes M , we evaluate the candidate mixtures using a heuristic objective function:

$$Score = \left| \frac{c_1^{cold}}{c_1^{hot}} \right|^2 \cdot |D_{hot} - D_{cold}|. \quad (18)$$

This scoring formula acts as a balance between two fundamental requirements for a macroscopic QME. The first multiplier (the ratio of squared projections) enforces the spectral condition: the 'cold' state must strongly excite the slowest relaxation channel, while the 'hot' state remains essentially 'dark' to it. The second multiplier acts as a crucial geometric regularizer. It ensures that the selected states start sufficiently far apart in the relative entropy metric space ($\Delta D(0) > 0$), preventing the algorithm from converging to trivially close states that would yield physically meaningless, instantaneous micro-crossings.

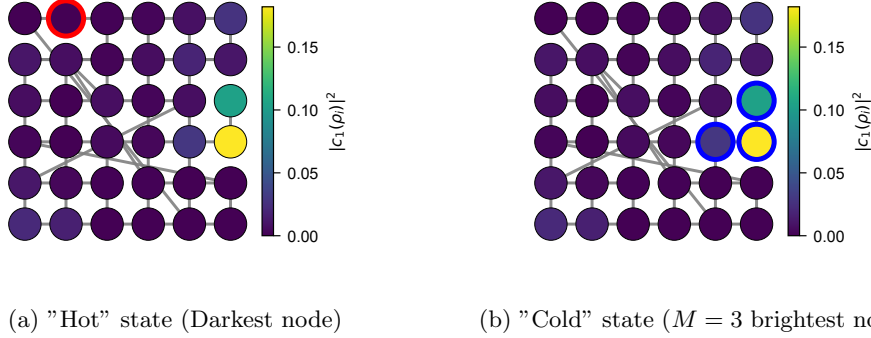


Fig. 12: Deterministic selection of the Mpemba pair based on the $k = 1$ excitability map from Fig. 11. **(A)** The localized "hot" state is placed at the darkest node (red outline), minimizing its interaction with the slow mode. **(B)** The mixed "cold" state is distributed across the $M = 3$ brightest nodes (blue outlines), identified as the optimal combination maximizing the heuristic score.

8.4 Monte Carlo Benchmark and Applicability Limits

Defining a universal, cross-system metric for the 'strength' of the QME remains a non-trivial challenge in open quantum systems, since absolute trajectory crossing times (t^*) are highly sensitive to the initial distance gap between states. Since evaluating

absolute optimality is hindered by this dependence, we evaluated the efficacy of the proposed deterministic algorithm by benchmarking it against a random Monte Carlo search within the fixed 6×6 network topology.

To ensure a physically meaningful comparison, we generated an ensemble of random pairs of diagonal density matrices whose populations were sampled from a uniform Dirichlet distribution, $p \sim \text{Dirichlet}(1, \dots, 1)$, ensuring unbiased coverage of the probability simplex. To discard trivial micro-crossings between states starting infinitesimally close, we applied a gap filter: the initial relative-entropy gap of a pair was required to be at least 10% of the gap engineered by our algorithm, yielding an ensemble of 30 000 admissible pairs.

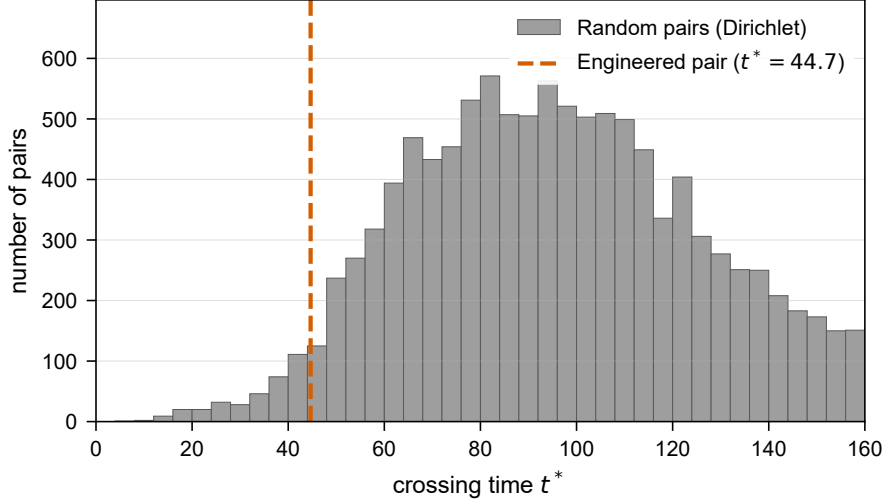


Fig. 13: Statistical validation of the deterministic algorithm. The histogram shows the distribution of crossing times t^* for an ensemble of 30 000 admissible random pairs (Dirichlet-sampled diagonal states) on the 6×6 network ($p = 0.15$). The vertical dashed line marks the crossing time ($t^* = 44.7$) of the deterministically engineered pair from Fig. 12: only 1.2% of the admissible pairs cross faster. Crossings beyond $t \approx 160$, where both trajectories have relaxed to within the numerical resolution of equilibrium, are not physically resolved and are omitted.

As illustrated in Fig. 13, the deterministically engineered pair significantly outperforms the random search: only about 1.2% of the admissible random pairs achieve a faster crossing. To characterize the strength of the effect beyond the crossing time, we also evaluated, for the same ensemble, the maximal subsequent advantage $\Delta D_{\max} \equiv \max_t [D_{\text{cold}}(t) - D_{\text{hot}}(t)]$. The engineered pair crosses late in the relaxation, at $D(t^*) \approx 3.6 \times 10^{-3}$, and reaches $\Delta D_{\max} \approx 2.4 \times 10^{-4}$; these absolute values are necessarily small, since by the crossing time both trajectories are already close to equilibrium. Relative to the ensemble, however, the advantage is exceptional: only 0.14% of the admissible random pairs achieve a larger $\Delta D_{\max}$ (the 99.86th percentile), while

the median ensemble advantage lies at the numerical noise floor ($\sim 10^{-15}$), i.e., a typical admissible pair never overtakes at all.

Finally, we note the physical limits of applicability for the proposed proof-of-concept heuristic. The current algorithm relies exclusively on the spatial excitability structure of the single fundamental mode v_1 . Consequently, its validity is contingent upon the network possessing a strongly localized slowest mode that is separated from the rest of the spectrum by a distinct eigenvalue gap. In arbitrary random ensembles, topological fluctuations can lead to quasi-degeneracies ($\lambda_1 \approx \lambda_2$) or delocalization of the fundamental mode, which would necessitate accounting for higher-order relaxation channels. The generalization of this geometric approach to multi-mode heuristics across broad topological ensembles remains a subject for future investigation.

9 Conclusion

We have presented a systematic study of the Quantum Mpemba effect in networks of coupled optical waveguides, progressing from regular square lattices to topologically disordered graphs.

For regular lattices, the spectral decomposition of the Liouvillian superoperator reveals that the relaxation eigenmodes possess well-defined macroscopic geometric patterns dictated by lattice symmetries. Exploiting this spatial regularity, we constructed a set of candidate initial states with highly selective mode content and confirmed that specific pairs of these states exhibit trajectory crossings in the relative-entropy distance to equilibrium—the hallmark of the Mpemba effect. The effect was observed robustly for square lattices of sizes $n = 5$ through $n = 11$.

Introducing topological disorder via a Watts–Strogatz-type edge-rewiring procedure, we tracked the evolution of the relaxation spectrum across the lattice-to-random crossover. Three key findings emerged. First, the relaxation rate of the slowest mode $\text{Re } \lambda_1$ decreases in absolute value with increasing rewiring probability beyond the small-world threshold $p \approx 0.005$, indicating that certain disordered configurations host anomalously slow relaxation channels—a spectral phenomenon we identify as topology-induced slow-mode localization. Second, the Inverse Participation Ratio and the community-overlap analysis demonstrate that the slowest mode undergoes a pronounced spatial localization, concentrating within a small subset of network nodes, while the second and third modes remain comparatively delocalized. Third, this localization asymmetry enables a natural classification of modes into “bright” (globally excitable) and “dark” (geometrically decoupled from most sites), providing a direct geometric route to engineering initial states with suppressed overlap with the slowest relaxation channel.

Building on these observations, we formulated a deterministic heuristic algorithm for constructing pairs of initial diagonal density matrices that maximize the Mpemba effect on a given disordered network. The algorithm places the “hot” (far-from-equilibrium, fast-relaxing) state at the node with the smallest local excitability for the first mode, and distributes the “cold” (closer-to-equilibrium, slow-relaxing) state across the M brightest nodes, with M optimized via a scoring function that balances

the projection ratio and the initial distance gap. A benchmark against 30 000 admissible random diagonal-state pairs on a 6×6 lattice at $p = 0.15$ confirmed that only about 1.2% of them achieve a faster crossing than the algorithmically engineered pair, validating the approach as a proof of concept.

Several directions for future work merit attention. The present analysis is restricted to incoherent (diagonal) initial states; extending the algorithm to coherent superpositions may further amplify the effect by exploiting off-diagonal interference. Likewise, incorporating site-resolved photon losses (beyond the uniform attenuation that factors out of the normalized dynamics) would bring the model closer to realistic photonic hardware. Scaling the approach to larger networks ($n \gg 10$) will require efficient spectral methods or variational approximations, since, for a graph of N nodes, the Liouvillian acts on the N^2 -dimensional space of density matrices and is represented by an $N^2 \times N^2$ matrix, whose dense storage scales as N^4 . It would also be valuable to investigate alternative graph families—such as scale-free, hierarchical, or empirical real-world networks—to assess the universality of the topology-induced mode localization. Finally, the connection between the topology-induced Mpemba effect and Liouvillian exceptional points, recently highlighted in the context of trapped-ion experiments [10], deserves further theoretical exploration in the network setting. From an experimental perspective, the model maps directly onto photonic mesh lattices with controllable dephasing [13, 14], suggesting that the predicted effects are within reach of current integrated-photonic and fiber-loop platforms.

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Declarations

Conflict of interest. The authors declare no conflict of interest.

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